

Problem 11

If $1 < p < \infty$, determine the cases for which Minkowski's inequality

$$\left\| \sum_i f_i \right\|_p \leq \sum_i \|f_i\|_p$$

holds with equality. What changes when $p = 1$ or $p = \infty$?

Solution.

$$\begin{aligned} \left\| \sum_{i \in I} f_i \right\|_p^p &= \int_X \left| \sum_{i \in I} f_i(x) \right|^p d\mu \\ &= \int_X \left| \sum_{i \in I} f_i(x) \right|^{p-1} \left| \sum_{i \in I} f_i(x) \right| d\mu \\ &\leq \int_X \left| \sum_{i \in I} f_i(x) \right|^{p-1} \left(\sum_{i \in I} |f_i(x)| \right) d\mu \end{aligned}$$

Note that the last inequality, arises due to the triangle inequality of complex numbers:

$$\left| \sum_{i \in I} f_i(x) \right| \leq \sum_{i \in I} |f_i(x)|.$$

Equality in the triangle inequality only holds if $f_i(x) = \lambda_i(x)f(x)$ for $\lambda_i(x) \in [0, \infty)$ and $f : X \rightarrow S^1$. Thus if $f_i = \lambda_i f$ almost everywhere, where $\lambda_i(x) \in [0, \infty)$ and $f : X \rightarrow S^1$ equality holds. So suppose this is the case.

$$\begin{aligned}
\left\| \sum_{i \in I} f_i \right\|_p^p &= \int_X \left(\left| \sum_{i \in I} f_i(x) \right|^{p-1} \sum_{i \in I} |f_i(x)| \right) d\mu \\
&= \int_X \sum_{i \in I} |f_i(x)| \left| \sum_{j \in I} f_j(x) \right|^{p-1} d\mu \\
&= \sum_{i \in I} \int_X |f_i(x)| \left| \sum_{j \in I} f_j(x) \right|^{p-1} d\mu \quad (\text{Tonell's Theorem}) \\
&= \sum_{i \in I} \int_X |f_i(x)| \left| \left(\sum_{j \in I} f_j(x) \right)^{p-1} \right| d\mu \\
&= \sum_{i \in I} \left\| f_i(x) \left(\sum_{j \in I} f_j(x) \right)^{p-1} \right\|_1 \\
&\leq \sum_{i \in I} \|f_i(x)\|_p \left\| \sum_{j \in I} f_j(x) \right\|_{\frac{p}{p-1}}
\end{aligned}$$

The final inequality comes from Holder.

Problem 12

Show that Hölder's inequality

$$\|fg\|_r \leq \|f\|_p \|g\|_q, \quad \frac{1}{p} + \frac{1}{q} = \frac{1}{r}$$

is equivalent to the log-convexity of L^p norms: for $0 < p < q \leq \infty$, $0 < \theta < 1$, and $\frac{1}{r} = \frac{1-\theta}{p} + \frac{\theta}{q}$,

$$\|f\|_r \leq \|f\|_p^{1-\theta} \|f\|_q^\theta.$$

Solution. $\Rightarrow$: Suppose Hölder's inequality is true for all choices of parameters. Suppose

$$\frac{1}{r} = \frac{1-\theta}{p} + \frac{\theta}{q}.$$

Let

$$\begin{aligned}
\alpha &= \frac{p}{1-\theta} \\
\beta &= \frac{q}{\theta}.
\end{aligned}$$

Thus by Hölder's inequality we have

$$\begin{aligned}
\|f\|_r &= \|f^{1-\theta} f^\theta\|_r \\
&\leq \|f^{1-\theta}\|_\alpha \|f^\theta\|_\beta \\
&= \left(\int_X |f(x)|^{(1-\theta)\alpha} d\mu \right)^{\frac{1}{\alpha}} \left(\int_X |f(x)|^{\theta\beta} d\mu \right)^{\frac{1}{\beta}} \\
&= \left(\int_X |f(x)|^p d\mu \right)^{\frac{1-\theta}{p}} \left(\int_X |f(x)|^q d\mu \right)^{\frac{\theta}{q}} = \|f\|_p^{1-\theta} \|f\|_q^\theta.
\end{aligned}$$

$\Leftarrow$: Suppose the log-convexity of L^p is true. Let

$$\frac{1}{r} = \frac{1}{p} + \frac{1}{q}.$$

We will first show that it is sufficient to prove that if $\|f\|_p \leq 1$ and $\|g\|_q \leq 1$ then $\|fg\|_r \leq 1$. WLOG $\|F\|_p \neq 0 \neq \|G\|_q$ otherwise F or G are zero almost everywhere and hence FG is zero almost everywhere and $\|FG\|_r = 0$ and the statement is trivially true. Let

$$\|FG\|_r \leq \|F\|_p \|G\|_q \left\| \left(\frac{F}{\|F\|_p} \right) \left(\frac{G}{\|G\|_q} \right) \right\|_r \leq \|F\|_p \|G\|_q.$$

Thus it is sufficient to prove that $\|f\|_p \leq 1$ and $\|g\|_q \leq 1$ implies that $\|fg\|_r \leq 1$. Consider the following function

$$F(x) := \begin{cases} \frac{f(x)}{|g(x)|^{\frac{q}{p}}} & \text{if } g(x) \neq 0 \\ 0 & \text{otherwise} \end{cases}.$$

Let $d\nu = |g(x)|^q d\mu$. Then we have that

$$\begin{aligned}
\|F\|_{L^p(X, d\nu)}^p &= \int_X F(x)^p d\nu \\
&= \int_{\{x: g(x) \neq 0\}} \left(\frac{|f(x)|^p}{|g(x)|^q} \right) |g(x)|^q d\mu \\
&= \int_{\{x: g(x) \neq 0\}} |f(x)|^p d\mu \\
&\leq \|f\|_{L^p(X, d\mu)}^p \leq 1.
\end{aligned}$$

We also have

$$\begin{aligned}
\|F\|_{L^r(X, d\nu)}^r &= \int_X F(x)^r d\nu \\
&= \int_{\{x: g(x) \neq 0\}} \frac{|f(x)|^r}{|g(x)|^{\frac{rq}{p}}} d\mu
\end{aligned}$$

Problem 13

Differentiate $\log \|f\|_p$ twice with respect to $\alpha := \frac{1}{p}$ and show that this is non-negative (take f to be a non-zero simple function with finite measure support, to avoid technicalities).

Solution.

Problem 18

On a measure space consisting of N points (with counting measure), and $0 < p \leq q \leq \infty$, one has

$$\|f\|_{\ell^q} \leq \|f\|_{\ell^p} \leq N^{\frac{1}{p}-\frac{1}{q}} \|f\|_{\ell^q}.$$

When does equality occur for either of these two inequalities? Note how the example that attains the lower bound is in many ways the “opposite extreme” to the example which attains the upper bound.

Solution.

Problem 19

Establish the (vector-valued) bound

$$\left\| \left(\sum_n |f_n|^q \right)^{\frac{1}{q}} \right\|_{L^p} \leq \left(\sum_n \|f_n\|_{L^p}^q \right)^{\frac{1}{q}}$$

for any measurable functions f_n and any $0 < q < p \leq \infty$.

Solution.